DOCTORAL CONSORTIUM · ASSETS 2026

*Designing **Autonomy-Preserving** Cognitive Support*

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Why autonomy matters

Autonomy is a basic psychological need, essential for well-being. Some people need more support with cognitive skills such as memory, attention and planning.

Technology can provide cognitive support at scale. But it is mostly designed to efficiently finish the task, so it does the task for people and not with them which undermines autonomy.

This affects most those who need support the most.

Who does the task?

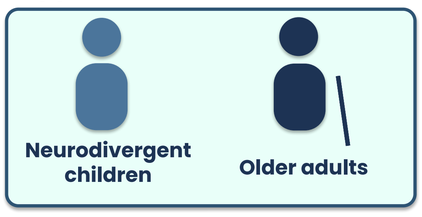
Doing for

Technology does it

Supporting

Help | **The person does it**

Who relies on cognitive support the most?



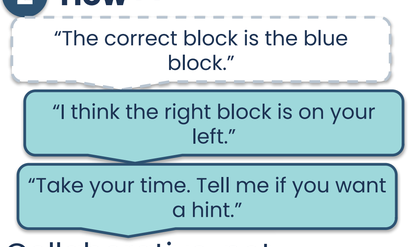
Autonomy support in prior work

1 How much [1]



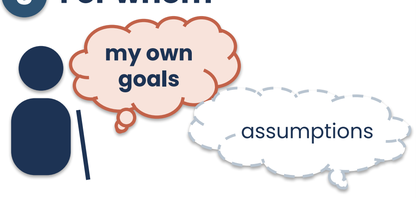
Fits what the person can still do, compensating for what they cannot.

2 How [2]



Collaborative, not imposed.
Offering choice, rationale

3 For whom



Designed for their own goals, not assumptions.

The gap. These considerations come from small studies in different domains and almost none directly measured the user's sense of autonomy. How to design for autonomy is still unclear.
To our knowledge, no published work has deployed a robot for cognitive support with autonomy as an explicit goal.

[1] Wood, Bruner & Ross (1976). [2] Ryan & Deci (2000).

How should **cognitive support be designed** to preserve users' sense of **autonomy**, and how can **assistive robots** contribute?

RQ1 How to estimate cognitive state in real time from what users do in a task?

Expected contribution: behavioral indicators of cognitive state.

RQ2 Design robot behaviors that preserve autonomy

Expected contribution: a repertoire of autonomy-preserving robot behaviors.

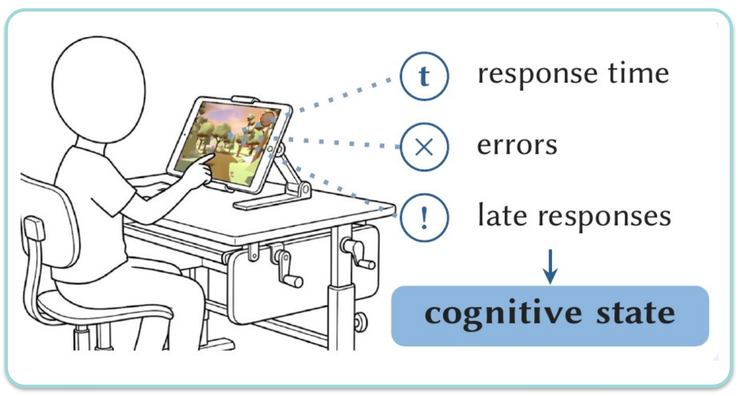
RQ3 Adapt when and how the robot supports

Expected contribution: a robot that provides adaptive cognitive support, and evidence on its effect on autonomy and performance.

Estimating cognitive state

Under review, CHI 2027

81 neurodivergent children, ages 7–14



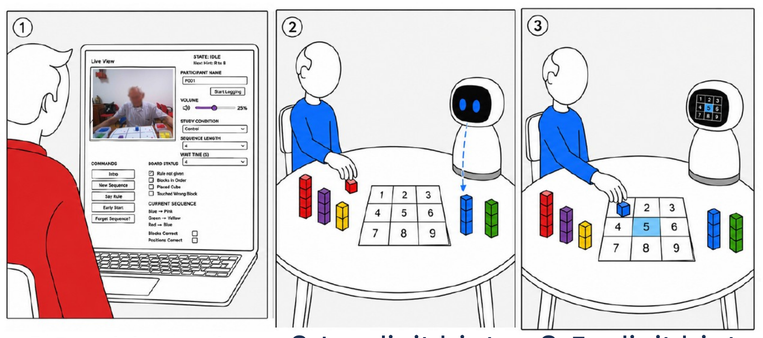
- Four gamified standardized tasks for a different cognitive skill: memory, attention, inhibitory control, cognitive flexibility.
- Three of four aligned with their standardized task at some levels.
- Features beyond the score were meaningful indicators of cognitive skill.

- **Contribution**
- Behavioral indicators of cognitive state in these games.
- Validation methodology that can be reused for further indicators.

Autonomy-preserving behaviors

Under review, HRI 2027

35 older adults, ages 64–93



1 Dashboard **2** Implicit hint **3** Explicit hint

Perceived autonomy

4.61 Implicit hints

1.82 Explicit hints

- Memory task where the participants had to remember an increasing sequence of colors and positions
- A robot gave hints, from subtle (a glance) to direct (the answer).
- Implicit hints gave higher perceived autonomy, with no cost to performance.

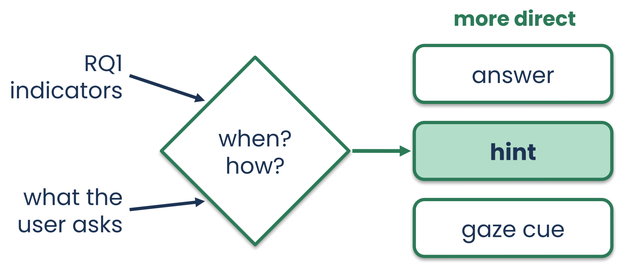
Contribution

- Early evidence of partial but coherent support that implicit support preserves autonomy.

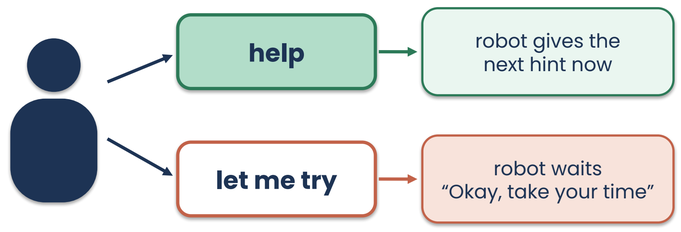
Adaptive support

Next steps

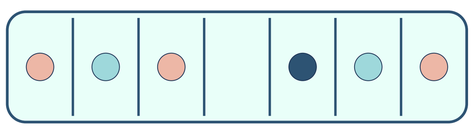
Adaptive robot. Tests whether adapting the support is better than fixed implicit support.



Offering choice. Tests whether the design assumptions about what people need match what they actually need.



Task. Real-world task, co-designed with older adults, carers and psychologists. For instance, medication management or grocery shopping.



Medication

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